

Timing-driven placement

Timing-driven place weights critical nets in the wirelength objective

The idea

- Multiply each net's HPWL by its criticality weight and sum
- Heavy weights pull critical nets shorter even when plain HPWL looks fine
- Always report both plain and weighted totals so you can see what the objective actually

Pseudocode

- Timing-driven place multiplies each net's HPWL by criticality and sums
- Pseudocode always reports plain and weighted totals together
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Starter plain fifty-two becomes timing one hundred sixteen because the four-pin net has
- Compact golden drops timing cost to thirty

Algorithm sketch — try these

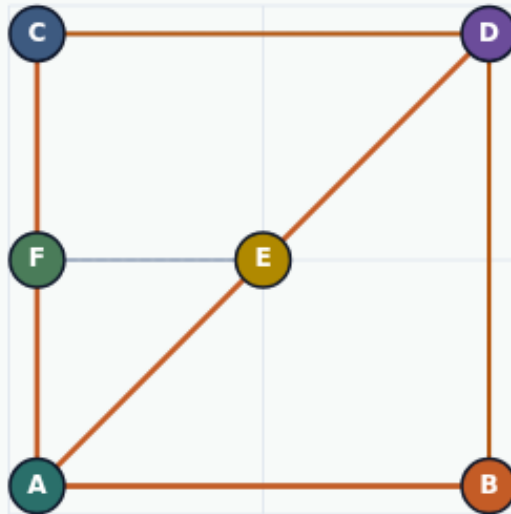
```
INPUT: positions, nets, weights w[net]
OUTPUT: plain HPWL, timing HPWL
plain  $\leftarrow \sum \text{HPWL}(\text{net})$ 
timing  $\leftarrow \sum w[\text{net}] \cdot \text{HPWL}(\text{net})$ 
optimize timing (or report both)
GOLDEN starter: plain=52 timing=116
compact: plain=14 timing=30
```

Plain HPWL hides critical nets

TIMING-DRIVEN PLACE

Plain HPWL hides critical nets

Step 1 / 5 · plain-vs-weighted · module04-01-timing-driven-place



Explanation

On the starter, plain HPWL is fifty-two, but timing-weighted HPWL is one hundred sixteen because the four-pin net carries weight five.

- weights = [1,1,1,1,5,1]
- Net ABCD is critical
- Report both totals

Plain HPWL: 52
Timing HPWL: 116
Net4 weight: 5

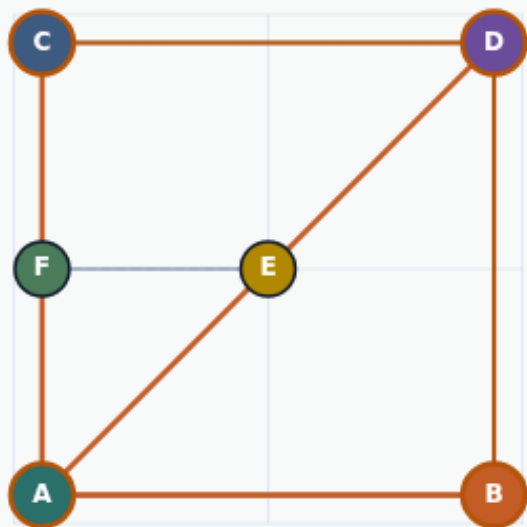
Plain HPWL hides critical nets

Weighted sum of net HPWLs

TIMING-DRIVEN PLACE

Weighted sum of net HPWLs

Step 2 / 5 · weight-math · module04-01-timing-driven-place



Explanation

Multiply each net's bbox HPWL by its criticality and sum. The critical ABCD net alone contributes five times sixteen equals eighty on the starter.

- $\sum w_i \cdot \text{HPWL}(\text{net}_i)$
- Heavy nets pull harder
- Same coordinates, new objective

ABCD plain: 16
ABCD weighted: 80
Total timing: 116

Weighted sum of net HPWLs

Golden timing cost drops to thirty

TIMING-DRIVEN PLACE

Golden timing cost drops to thirty

Step 3 / 5 · golden-timing · module04-01-timing-driven-place



Explanation

The compact golden cuts the critical bbox sharply. Timing-weighted HPWL falls from one hundred sixteen to thirty while plain HPWL hits fourteen.

- Starter timing 116 → golden 30
- Plain golden still 14
- Critical net drove the win

```
goldenTimingHpwl: 30
goldenHpwl: 14
Measured timing: 30
```

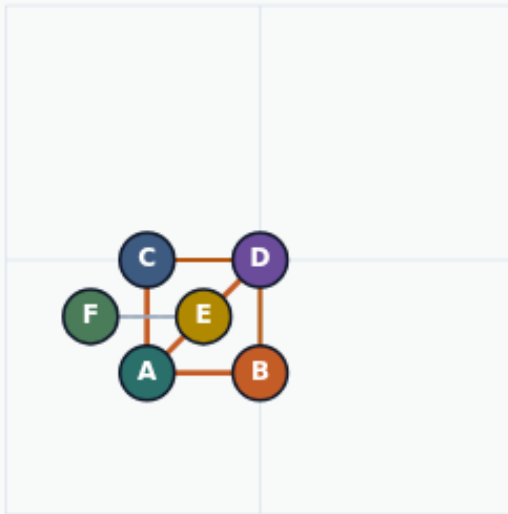
Golden timing cost drops to thirty

Always quote plain and weighted

TIMING-DRIVEN PLACE

Always quote plain and weighted

Step 4 / 5 · both-reports · module04-01-timing-driven-place



Explanation

A placement can look fine on plain HPWL while the critical net stays long. Timing labs demand both numbers so the objective is visible.

- Plain: wirelength yardstick
- Weighted: timing objective
- Critical net highlighted

starter: 52 / 116
golden: 14 / 30

Always quote plain and weighted

Weights change what you optimize

TIMING-DRIVEN PLACE

Weights change what you optimize

Step 5 / 5 · takeaway · module04-01-timing-driven-place



Explanation

Timing-driven place is still wirelength—just weighted. Remember one hundred sixteen to thirty on this instance, and never drop the plain HPWL report.

- Critical net weight 5
- 116 → 30 timing HPWL
- Course wrap: compare all engines

```
starterTimingHpwl: 116  
goldenTimingHpwl: 30
```

Weights change what you optimize

Browser lab track

- In the browser lab track, open the **timing-driven-place** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse ``tiny_place.json``, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module